

- Datatype conversion can happen either explicitly or implicitly.
- Operators are constructs that manipulate the value of operands. Operators may be unary or binary.
- An expression is a combination of values, variables and operators.
- Python has `input()` function for taking user input.
- Python has `print()` function to output data to a standard output device.

NOTES

EXERCISE

1. Which of the following identifier names are invalid and why?

i	Serial_no.	v	Total_Marks
ii	1st_Room	vi	total-Marks
iii	Hundred\$	vii	_Percentage
iv	Total Marks	viii	True

2. Write the corresponding Python assignment statements:
 - a) Assign 10 to variable `length` and 20 to variable `breadth`.
 - b) Assign the average of values of variables `length` and `breadth` to a variable `sum`.
 - c) Assign a list containing strings 'Paper', 'Gel Pen', and 'Eraser' to a variable `stationery`.
 - d) Assign the strings 'Mohandas', 'Karamchand', and 'Gandhi' to variables `first`, `middle` and `last`.
 - e) Assign the concatenated value of string variables `first`, `middle` and `last` to variable `fullname`. Make sure to incorporate blank spaces appropriately between different parts of names.
3. Write logical expressions corresponding to the following statements in Python and evaluate the expressions (assuming variables `num1`, `num2`, `num3`, `first`, `middle`, `last` are already having meaningful values):
 - a) The sum of 20 and -10 is less than 12.
 - b) `num3` is not more than 24.

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- c) 6.75 is between the values of integers num1 and num2.
 - d) The string 'middle' is larger than the string 'first' and smaller than the string 'last'.
 - e) List Stationery is empty.
4. Add a pair of parentheses to each expression so that it evaluates to True.
- a) `0 == 1 == 2`
 - b) `2 + 3 == 4 + 5 == 7`
 - c) `1 < -1 == 3 > 4`
5. Write the output of the following:
- a)

```
num1 = 4
num2 = num1 + 1
num1 = 2
print (num1, num2)
```
 - b)

```
num1, num2 = 2, 6
num1, num2 = num2, num1 + 2
print (num1, num2)
```
 - c)

```
num1, num2 = 2, 3
num3, num2 = num1, num3 + 1
print (num1, num2, num3)
```
6. Which data type will be used to represent the following data values and why?
- a) Number of months in a year
 - b) Resident of Delhi or not
 - c) Mobile number
 - d) Pocket money
 - e) Volume of a sphere
 - f) Perimeter of a square
 - g) Name of the student
 - h) Address of the student
7. Give the output of the following when num1 = 4, num2 = 3, num3 = 2
- a)

```
num1 += num2 + num3
print (num1)
```
 - b)

```
num1 = num1 ** (num2 + num3)
print (num1)
```
 - c)

```
num1 **= num2 + num3
```
 - d)

```
num1 = '5' + '5'
print(num1)
```

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- e) `print(4.00/(2.0+2.0))`
- f) `num1 = 2+9*((3*12)-8)/10`
`print(num1)`
- g) `num1 = 24 // 4 // 2`
`print(num1)`
- h) `num1 = float(10)`
`print (num1)`
- i) `num1 = int('3.14')`
`print (num1)`
- j) `print('Bye' == 'BYE')`
- k) `print(10 != 9 and 20 >= 20)`
- l) `print(10 + 6 * 2 ** 2 != 9//4 -3 and 29`
`>= 29/9)`
- m) `print(5 % 10 + 10 < 50 and 29 <= 29)`
- n) `print((0 < 6) or (not(10 == 6) and`
`(10<0)))`

8. Categorise the following as syntax error, logical error or runtime error:

- a) `25 / 0`
- b) `num1 = 25; num2 = 0; num1/num2`

9. A dartboard of radius 10 units and the wall it is hanging on are represented using a two-dimensional coordinate system, with the board's center at coordinate (0,0). Variables x and y store the x-coordinate and the y-coordinate of a dart that hits the dartboard. Write a Python expression using variables x and y that evaluates to True if the dart hits (is within) the dartboard, and then evaluate the expression for these dart coordinates:

- a) (0,0)
- b) (10,10)
- c) (6, 6)
- d) (7,8)

10. Write a Python program to convert temperature in degree Celsius to degree Fahrenheit. If water boils at 100 degree C and freezes as 0 degree C, use the program to find out what is the boiling point and freezing point of water on the Fahrenheit scale.

(Hint: $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 9/5 + 32$)

11. Write a Python program to calculate the amount payable if money has been lent on simple interest.

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Principal or money lent = P, Rate of interest = R% per annum and Time = T years. Then Simple Interest (SI) = $(P \times R \times T) / 100$.

Amount payable = Principal + SI.

P, R and T are given as input to the program.

12. Write a program to calculate in how many days a work will be completed by three persons A, B and C together. A, B, C take x days, y days and z days respectively to do the job alone. The formula to calculate the number of days if they work together is $xyz / (xy + yz + xz)$ days where x, y, and z are given as input to the program.
13. Write a program to enter two integers and perform all arithmetic operations on them.
14. Write a program to swap two numbers using a third variable.
15. Write a program to swap two numbers without using a third variable.
16. Write a program to repeat the string "GOOD MORNING" n times. Here 'n' is an integer entered by the user.
17. Write a program to find average of three numbers.
18. The volume of a sphere with radius r is $4/3\pi r^3$. Write a Python program to find the volume of spheres with radius 7cm, 12cm, 16cm, respectively.
19. Write a program that asks the user to enter their name and age. Print a message addressed to the user that tells the user the year in which they will turn 100 years old.
20. The formula $E = mc^2$ states that the equivalent energy (E) can be calculated as the mass (m) multiplied by the speed of light ($c = \text{about } 3 \times 10^8 \text{ m/s}$) squared. Write a program that accepts the mass of an object and determines its energy.
21. Presume that a ladder is put upright against a wall. Let variables length and angle store the length of the ladder and the angle that it forms with the ground as it leans against the wall. Write a Python program to compute

the height reached by the ladder on the wall for the following values of length and angle:

- a) 16 feet and 75 degrees
- b) 20 feet and 0 degrees
- c) 24 feet and 45 degrees
- d) 24 feet and 80 degrees

CASE STUDY-BASED QUESTION

Schools use “Student Management Information System” (SMIS) to manage student-related data. This system provides facilities for:

- recording and maintaining personal details of students.
- maintaining marks scored in assessments and computing results of students.
- keeping track of student attendance.
- managing many other student-related data. Let us automate this process step by step.

Identify the personal details of students from your school identity card and write a program to accept these details for all students of your school and display them in the following format.

Name of School	
Student Name: PQR	Roll No: 99
Class: XI	Section: A
Address : Address Line 1	
Address Line 2	
City: ABC	Pin Code: 999999
Parent's/ Guardian's Contact No: 9999999999	

DOCUMENTATION TIPS

It is a fact that a properly documented program is easy to read, understand and is flexible for future development. Therefore, it is important that one pays extra attention to documentation while coding. Let us assess the documentation done by us in our case study program and also find out whether our friends also pay similar attention to documentation or not.

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Following is a checklist of good documentation points:

- Objective of the program is clearly stated in the beginning.
- Objective of each function is clearly mentioned in the beginning of each function.
- Comments are inserted at the proper place so as to enhance the understandability and readability of the program.

(Note: Over commenting doesn't help)

- Variables and function names are meaningful and appropriate.
- Single letter variable names are not used.
- Program name is meaningful.

(Note: It is not proper to use your name as program name, for example, 'raman.py' or 'namya.py' to denote your program code. It is more appropriate to use the program name as 'bankingProject.py' for a banking related program or 'admProcess' for an admission related program.)

- Program code is properly indented.
- Same naming conventions are used throughout the program.

(Note: Some of the naming conventions are firstNum, first_num, to denote the variable having first number).

Let's do this exercise for our peer's case studies as well and provide a feedback to them.

A relevant peer feedback helps in improving the documentation of the projects. It also helps in identifying our mistakes and enriches us with better ideas used by others.